4.1. INTRODUCTION

Generally every piece of electronics equipment in the world today is powered from a DC Source. This source may be either a battery or a power supply. Most electronics equipment requires not only a DC power source, but one that is well filtered and well regulated as well.

Three types of electronic power conversion devices are in common use today: the AC to DC Supply, the DC to DC Converter, and the DC to AC inverter.

DO YOU KNOW?

What is the other name for Power supply

Power supply is also called as "Power Source"

4.1.1 AC to DC Power Supply

Converter

It is a device that converts alternative current into direct current.

Applications

1. Radio and television receiver use this type of power supply.
2. It is also used in DVD player.
3. AC to DC power supply is used in all type of amplifiers.

4.1.2 DC to DC Converter

It is a device that converts a source of direct current (DC) from one voltage level to another.

Application

It is used in cellular phones and laptop computers.

4.1.3 DC to AC inverter

It is a device that converts a source of direct current into alternative current. It is also called inverter.

Application

An uninterruptible power supply uses an inverter to supply AC power when the main power is not available.

Each has its own specific areas of application. Of the three, AC to DC power supply is the most commonly used power supply. The objective of this lesson is how to rectify AC into DC power supply.

4.2. POWER SUPPLY BASICS

Now a days almost all electronics equipment consist a circuit that converts AC supply into DC supply is called DC Power Supply unit. A DC power supply unit (Commonly called as PSU) deriving power from the AC mains (Line). It performs a number of tasks.

The basic function of power supply is illustrated in Figure 4.1.

Transformer: Transformer is used to step down the AC supply voltage (220 V AC, 50 Hz) as per the requirement of the solid state electronic devices.

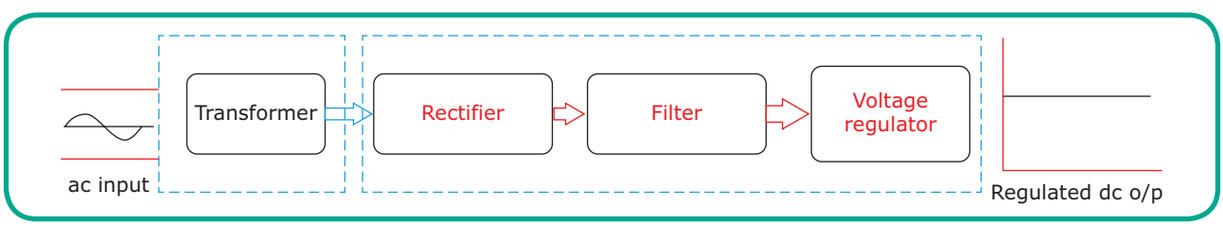


Figure 4.1 Basic power supply

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